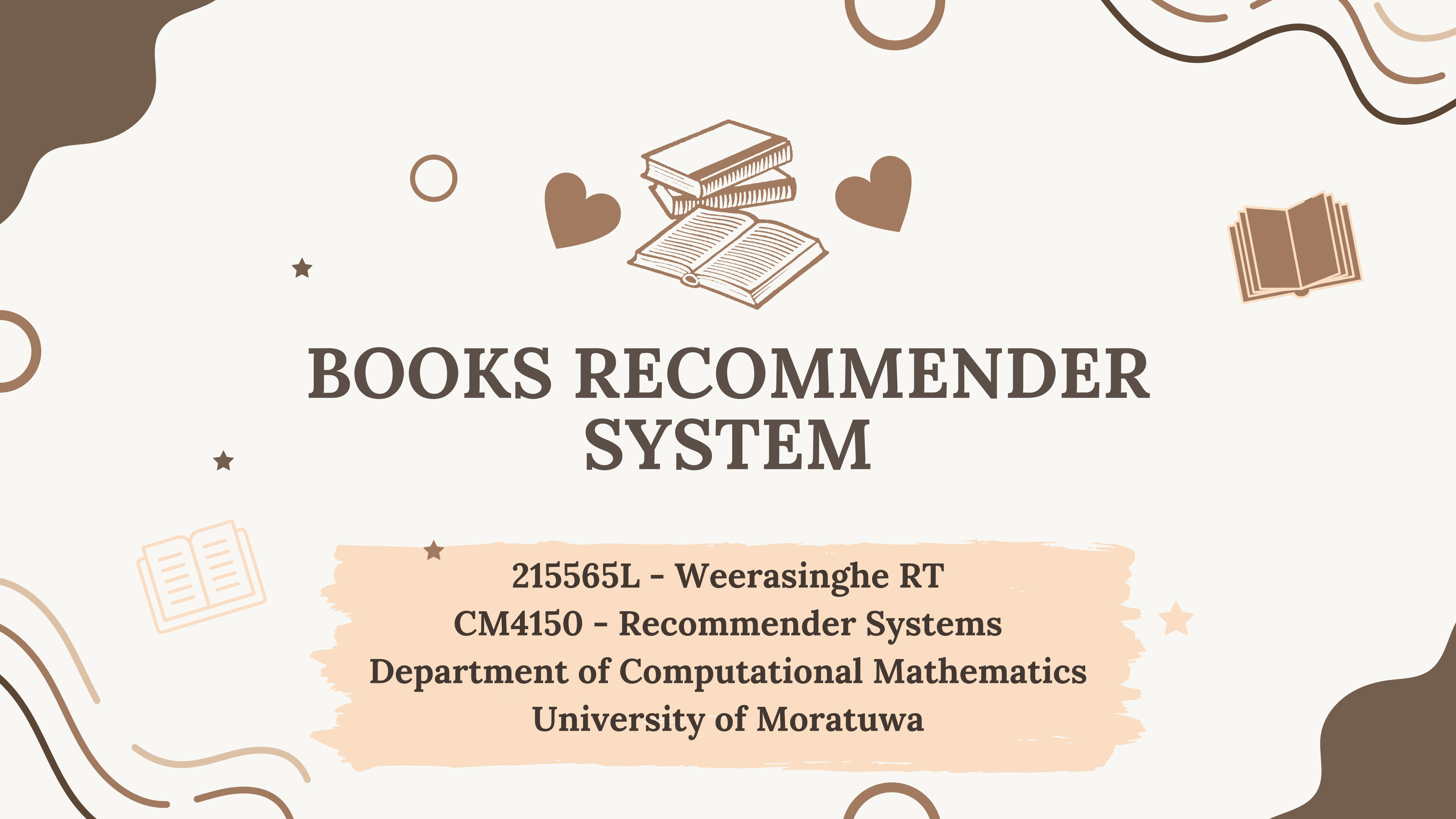




# BOOKS RECOMMENDER SYSTEM



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# PROBLEM

- › Finding a good book that matches your taste is difficult

## Why?

- › Every person has a different taste
- › New readers don't know what to read
- › Your taste changes over time
- › Book ratings differ with age





# SOLUTION

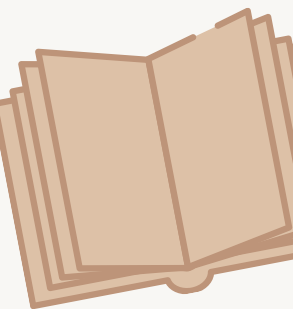
Hybrid book recommender system  
that addresses those concerns

## Overview

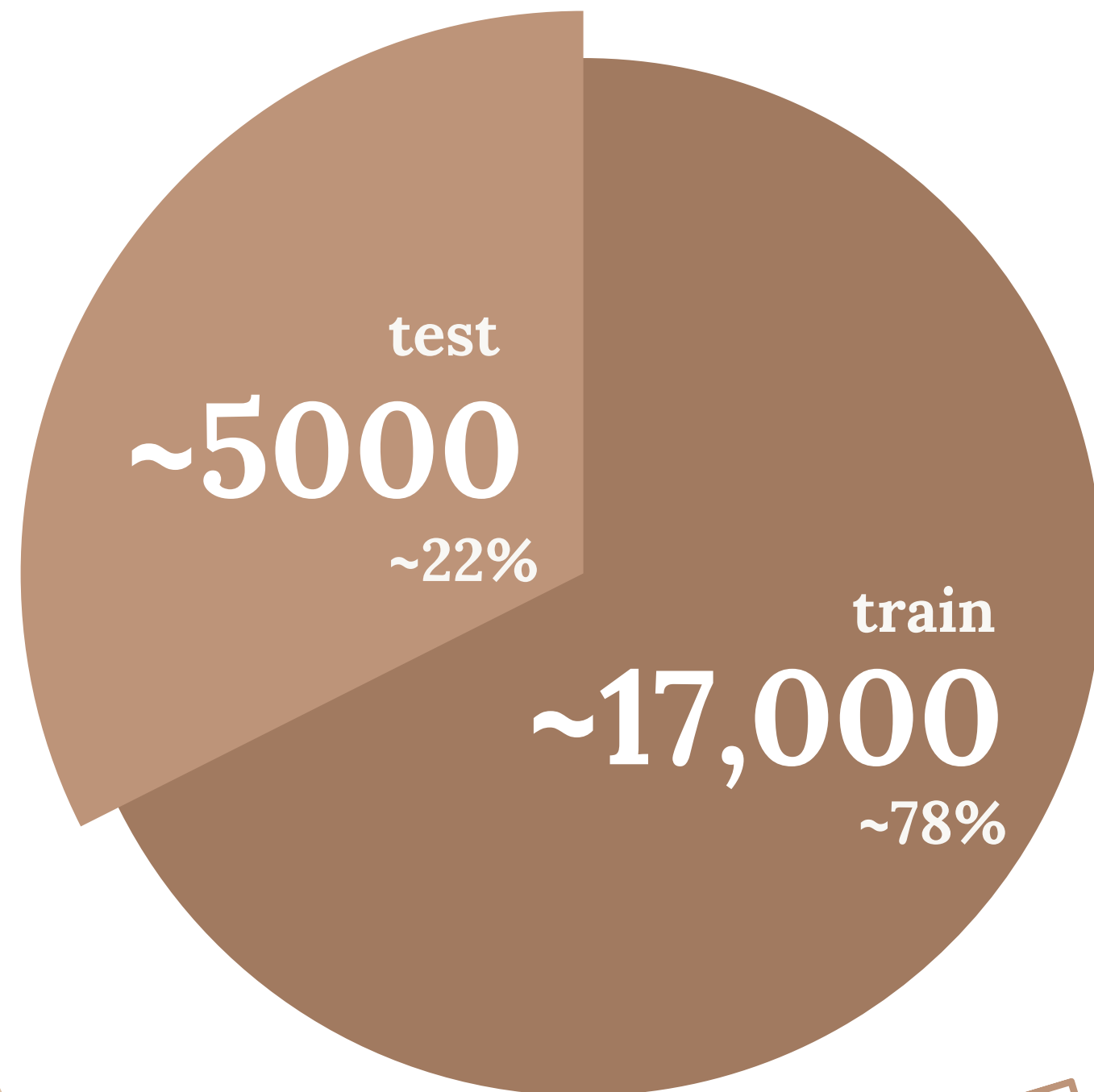
- Who → book readers (new, avid readers)
- What → books to read
- Objective → top-10 recommendations
- Data → Book Crossing Community

270K+ users, 270K+ books, 1M+ interactions

→ ~1K users, ~2K books, ~22K interactions



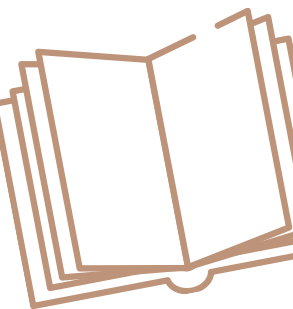
# DATA HANDLING



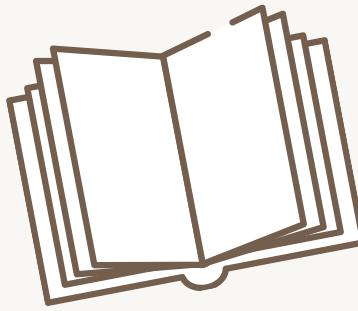
Interaction counts



- Clean & fix data issues (author, year)
- Feature preprocessing → TF-IDF (author, year, title)
- Train - test split



# ALGORITHMS



## Content based

## Collaborative

## Personalization

## Popularity recommendation

TF-IDF on book titles  
unigrams, bigrams  
500 max features  
> Your taste <

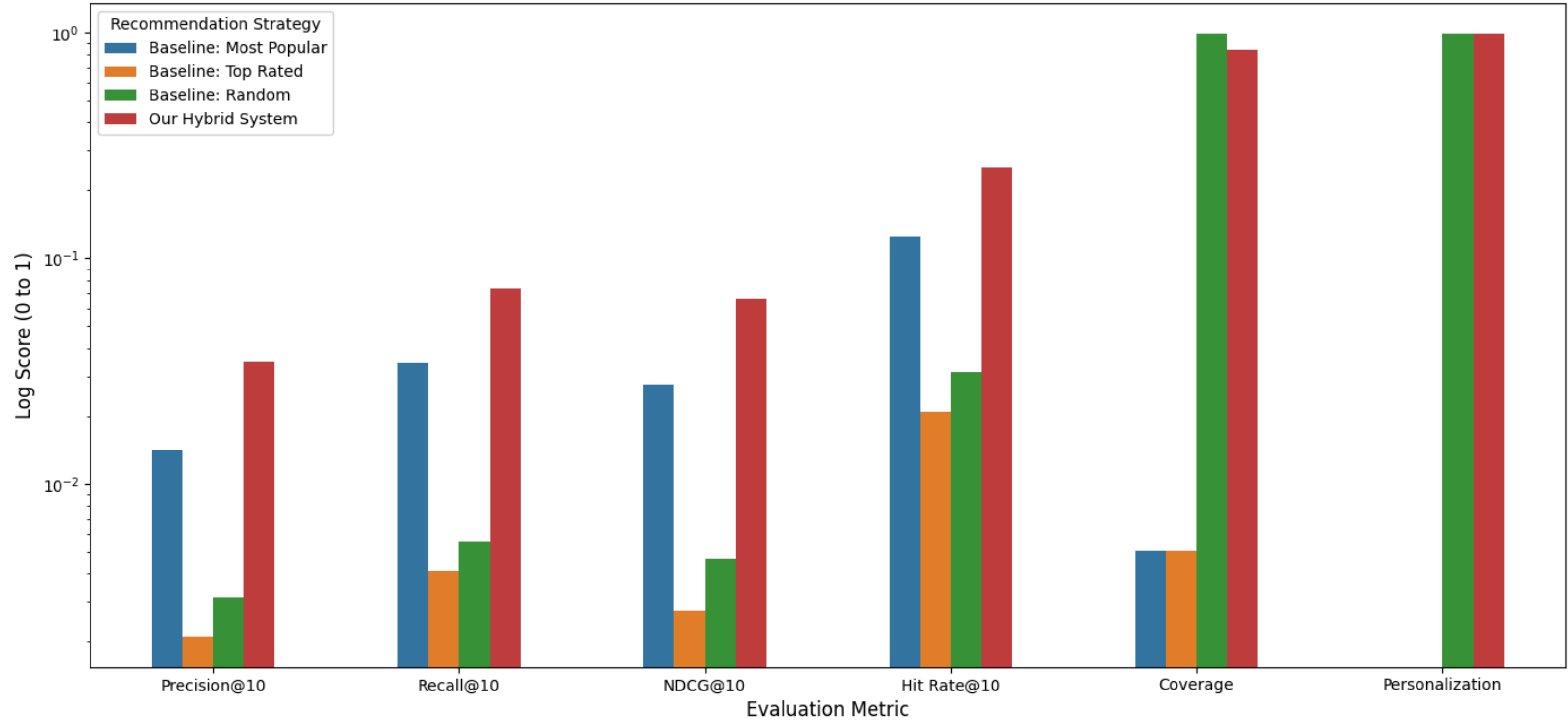
User CF  
Similar users  
Weighted ratings  
> General taste <

Adjust  
recommendations on  
your age & publication  
year

For limited history users,  
recommend popular  
books.  
Found with PageRank



# EVALUATIONS





**DEMO**